

The background features a close-up of green, elongated leaves, possibly from a plant like a bamboo or a similar species, with visible veins. A large, white, torn-paper-like shape is centered on the page, serving as a container for the text.

Plant Proposal – Tier 1

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The Problem

Plant disease is prevalent in 70% to 80% of the total plant population with mitigation being managed by integrated pest management. The destruction of 40% of global crops during harvest has cost the US agricultural industry \$220 billion annually.

The Solution

An automated mobile camera system would capture the leaf images in high resolution before processing them through the deep learning AI model to identify pathologies and the early stages of severity, which then delivers the mitigation steps so that crop loss and management costs are reduced.

Field Camera Photo > AI Model >
Diagnostic & Severity Rate + IPM
Guidelines

Technical Approach

Segmentation and Classification

Multi-Task Decoupled YOLO-Segmentation Architecture

Ultralytics PyTorch

The technical approach is done this way due to the varying factors when it comes to detecting disease in plants.

Data Plan

PlantVillage – Public Dataset

54,305 images divided into 38 different classes

[Species of Leaf] + [Health Status/Disease Type]

Success Metrics

Primary metric: 90% recall

Secondary metric: Latency of under 0.15 seconds per image.

Risks and Resources

Risk 1: Model degradation

Solution: Applying a background substitution and data augmentations before training.

Risk 2: Task Dominance

Solution: Implementing frameworks like Dynamic Weight Average.

Estimated cost: \$0

Your goal	Milestone	16-week term
Plan approved	Midterm submitted	Week 10
A pretrained model runs end to end on a few sample images	Something works, even if rough	Week 11
Add your data, your training if your tier needs it, and your application logic (counting, alerts, reports)	System works on YOUR problem	Weeks 12 to 13
Test, fix, and measure against your success metrics	Metrics recorded	Week 14
Demo video, README, final slides	Final submitted	Week 15

Milestone Plan